Comprehensive Methodology of Crop Rotation Analysis: Stochastic Processes, Econometrics, and Deep Learning Architectures

1. Introduction: The Complexity of Agricultural Sequence Modeling

The practice of crop rotation—the systematic sequencing of different crop species on the same land over time—serves as the agronomic backbone of sustainable agriculture. While the concept is ancient, dating back to the Roman *alimenta* and the medieval three-field system, the methodology for analyzing, quantifying, and predicting these sequences has undergone a radical transformation in the digital age. Contemporary rotation analysis is no longer merely a descriptive exercise in agronomy; it has evolved into a high-dimensional computational challenge that intersects stochastic probability theory, econometric utility maximization, and state-of-the-art deep learning.

The necessity for advanced analytical frameworks arises from the intricate web of variables influencing a farmer's decision to plant a specific crop. This decision is not an independent random event but a conditional outcome heavily weighted by biological history (e.g., soil pathogen load, nitrogen residue), economic constraints (e.g., commodity futures, fertilizer costs), and environmental stochasticity (e.g., drought patterns, climate change). Consequently, the data representing these decisions exhibits strong temporal autocorrelation, spatial dependency, and non-stationary dynamics.

To understand these systems, researchers have progressed from simple frequency tables to **Markov Chain** models that simulate state transitions, to **Multinomial Logit (MNL)** frameworks that explain economic causality, and finally to **Recurrent Neural Networks (RNNs)** and **Transformers** that capture long-range dependencies in "crop languages." This report provides an exhaustive technical examination of these methodologies. It deconstructs the mathematical formulations of transition probability matrices, evaluates the statistical rigor of mixed models in long-term experiments, and explores the paradigm shift towards attention-based neural architectures. By synthesizing diverse literature—from the statistical mechanics of the USDA's crop predictions to the neural attention maps of satellite imagery analysis—we establish a unified theory of crop rotation methodology.

2. Foundations of Data Acquisition and Representation

Before dissecting the analytical models, it is critical to understand the data infrastructure that powers them. The methodology of rotation analysis is fundamentally constrained by the

quality, resolution, and temporal depth of the underlying land-use data.

2.1 Remote Sensing and the Cropland Data Layer (CDL)

Modern analysis predominantly relies on remote sensing data, specifically the USDA National Agricultural Statistics Service (NASS) Cropland Data Layer (CDL). The CDL provides annual, georeferenced, crop-specific land cover data, typically at a 30-meter resolution. This dataset allows researchers to construct "pixel histories"—time-series vectors representing the sequence of crops grown on a specific plot of land over decades.[1]

However, the raw data requires rigorous preprocessing to be viable for high-order modeling:

- **Temporal Harmonization:** Crop rotation analysis requires consistent time steps. Researchers must account for cloud cover and sensor errors (e.g., Landsat 7 SLC-off gaps) using interpolation techniques or by integrating multi-sensor data (Sentinel-2, MODIS) to ensure a continuous timeline.[3]
- **Spatial Aggregation:** While pixel-level analysis offers granularity, it suffers from "salt-and-pepper" noise. Methodologies often involve aggregating pixels into fields or "Common Land Units" (CLUs) to derive a dominant crop type for a management unit, or using object-based image analysis (OBIA) to segment land parcels before classification.[4]

2.2 Symbolic Representation and Encoding

For computational models, crop types must be encoded into numerical formats.

- **One-Hot Encoding:** In traditional neural networks and Markov models, crops are often represented as orthogonal binary vectors (dummy variables). If there are k crop types, Corn might be $[1, 0, 0, \dots]$ and Soybean $[0, 1, 0, \dots]$. This creates sparsity issues in high-dimensional spaces.[1]
- **Symbolic Embeddings:** Advanced Transformer methodologies utilize "symbolic embeddings." Similar to how words are embedded in Natural Language Processing (NLP), crops are mapped to dense vector spaces where agronomic similarities are preserved (e.g., "Corn" and "Sorghum" might have a smaller Euclidean distance than "Corn" and "Alfalfa"). This approach allows models to generalize better across similar crop functional types.[5]

3. Stochastic Methodologies: The Markovian Paradigm

The most established methodology for analyzing crop rotation dynamics is the **Markov Chain**. This approach treats crop sequencing as a stochastic process where the future state is a probabilistic function of the current and past states.

3.1 First-Order Markov Chains and Transition Probability Matrices (TPM)

The foundational assumption of a first-order Markov Chain is the *Markov Property*, or "memorylessness." It posits that the probability of transitioning to crop state j at time $t+1$

depends solely on the crop state i at time t:

$$P(X_{t+1} = j \mid X_t = i, X_{t-1}, \dots, X_0) = P(X_{t+1} = j \mid X_t = i)$$
These probabilities are aggregated into a **Transition Probability Matrix (TPM)**, denoted as Π or P, where each element p_{ij} represents the conditional probability of moving from row i to column j.

3.1.1 Estimation of Transition Probabilities

The elements of the TPM are typically estimated using Maximum Likelihood Estimation (MLE) derived from historical land-use transitions. For a set of observed transitions n_{ij} (number of times crop i was followed by crop j):

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_{k} n_{ik}}$$
This empirical calculation provides a snapshot of the rotational "rules" of a region. For example, in the US Corn Belt, the transition matrix reveals the structural dominance of the corn-soybean rotation. Research extracting values from Kentucky and Iowa datasets indicates specific probabilistic signatures [6]:

Transition (t→t+1)	Probability Estimate	Agronomic Context
Corn $\to$ Corn	**0.411**	High persistence due to ethanol demand or continuous cropping on prime soils.
Corn $\to$ Soybean	**0.239 - 0.450**	Standard rotation to break pest cycles and fix nitrogen.
Soybean $\to$ Corn	**0.309 - 0.550**	Strong agronomic incentive; soybeans leave N-credit for corn.
Soybean $\to$ Soybean	**< 0.150**	Generally avoided due to disease pressure (e.g., nematodes), though values vary by price.

Table 1: Representative transition probabilities for major row crops. Values derived from aggregated datasets in.[6] Note that specific values fluctuate based on the years sampled and market conditions.

3.1.2 Stationarity and Equilibrium

A critical analytical step is testing for **stationarity**—whether the transition probabilities p_{ij} remain constant over time ($p_{ij}(t) = p_{ij}$). If stationarity holds, the system evolves towards a **Steady State Distribution** (π), calculated by solving the eigenvector equation:

$$\pi P = \pi \quad \text{subject to} \quad \sum \pi_i = 1$$

This vector π represents the long-term equilibrium proportion of the landscape occupied by each crop.[8] However, recent methodologies increasingly reject the stationarity assumption due to climate change and shifting policy landscapes (e.g., the Renewable Fuel Standard), necessitating the use of **Non-Homogeneous Markov Chains** where P varies as a function of time t or external covariates Z_t.[8]

3.2 High-Order Markov Chains (HOMC): Addressing Memory

The first-order assumption is often agronomically invalid. A farmer's decision in year t is frequently influenced by the crop in $t-2$ (e.g., to avoid planting corn three years in a row). To address this, methodologies employ **High-Order Markov Chains (HOMC)**.
An m-th order chain conditions the probability on the past m states:

$$P(X_{t+1} \mid X_t, X_{t-1}, \dots, X_{t-m+1})$$
While HOMCs improve agronomic realism, they introduce severe methodological hurdles:

1. **Parameter Explosion:** The number of parameters in the TPM scales as K^m, where K is the number of crop types and m is the order. For 10 crop types and a 4-year rotation memory, the matrix would require estimating 10^4 (10,000) transition probabilities.[1]
2. **Sparsity and Identifiability:** Most specific crop sequences (e.g., Cotton-Rice-Soybean-Corn) appear rarely in the data, leading to a sparse count matrix. This results in zero-probability estimates for transitions that are possible but unobserved, or high variance in estimates. Methodologies to mitigate this include **Variable Length Markov Chains (VLMC)** or mixture models that cluster similar rotation patterns to reduce dimensionality.[1]

3.3 Spatial Markov Chains

Standard Markov models assume spatial independence, treating each field as an isolated island. In reality, land use exhibits strong positive spatial autocorrelation due to shared soil characteristics, local climate, and neighborhood effects (social learning among farmers). **Spatial Markov Chains** extend the methodology by conditioning transition probabilities on the state of neighboring spatial units. This involves constructing a **Spatial Weight Matrix (W)**. The transition probability becomes:

$$P(X_{it+1} = j \mid X_{it} = k, WX_{it} \in L)$$
Where WX_{it} represents the spatial lag (the prevailing land use of neighbors) falling into category L. Studies utilizing this approach in the Corn Belt to model tillage adoption have shown that a field is significantly more likely to adopt conservation tillage if its neighbors have already done so, quantifying the "contagion" effect of agricultural practices.[7]

4. Econometric Approaches: Discrete Choice and

Utility Maximization

While Markov chains describe the *mechanics* of transition, they often fail to explain the *drivers*. Econometric methodologies, particularly **Discrete Choice Models (DCM)**, ground rotation analysis in microeconomic theory, viewing the crop sequence as a result of rational utility maximization.

4.1 The Random Utility Model (RUM) Framework

The cornerstone of this approach is the Random Utility Model. It posits that a farmer n chooses crop j from a set of alternatives C_n to maximize utility U_{nj}. The utility function is decomposed into a deterministic component V_{nj} and a stochastic error term ϵ_{nj}:

$$U_{nj} = V_{nj} + \epsilon_{nj} = \beta' x_{nj} + \gamma' z_n + \epsilon_{nj}$$

Here:

- x_{nj} represents attributes specific to the crop (e.g., expected price, input costs).
- z_n represents attributes specific to the farmer or land (e.g., soil quality, machinery ownership).
- **Rotation History:** Crucially, previous crops are included as dummy variables in x_{nj}. The coefficient β_{rot} associated with "Previous Crop = Corn" when "Current Choice = Soybean" captures the **rotation effect**—the explicit economic or yield benefit derived from the sequence.[6]

4.2 Multinomial Logit (MNL) Models

If the error terms ϵ_{nj} are assumed to be Independent and Identically Distributed (IID) following a Gumbel (Type I Extreme Value) distribution, the probability of choosing crop j takes the closed-form **Multinomial Logit** expression:

$$P_{nj} = \frac{\exp(\beta' x_{nj})}{\sum_{k \in C_n} \exp(\beta' x_{nk})}$$

This formulation allows researchers to quantify the "Marginal Effects" of variables. For instance, estimating how a 10% increase in fertilizer price alters the probability of rotating from Corn to Soybean.[6]

4.2.1 Dynamic Discrete Choice

A limitation of standard MNL is its static nature. Farmers are forward-looking agents. Dynamic Discrete Choice models (often solved via Dynamic Programming) treat the crop choice as an optimization problem over an infinite horizon.

$$V(S_t) = \max_{d_t} \{ u(S_t, d_t) + \delta E \}$$

Here, the value function V depends on the state S_t (which includes current soil fertility

and past crops). The farmer chooses decision d_t (current crop) to maximize immediate profit u plus the discounted (δ) expected future value. This methodology allows for the analysis of complex strategies, such as investing in a cover crop (which has negative immediate profit) to boost future yields.12

5. Statistical Frameworks for Long-Term Experiments (LTEs)

In contrast to observational studies using satellite data or survey records, **Long-Term Experiments (LTEs)** are controlled trials designed to validate the causal impacts of rotation. The methodology here shifts from prediction to rigorous hypothesis testing, primarily utilizing **Linear Mixed Models (LMM)**.

5.1 The Necessity of Mixed Models

Traditional Analysis of Variance (ANOVA) is often insufficient for LTEs due to:

1. **Unbalanced Data:** Long-term trials often suffer from missing data points over decades.
2. **Repeated Measures:** Observations on the same plot over time are correlated.
3. **Random Effects:** Blocks and years are best conceptualized as random samples from a larger population of environmental conditions.

The General Linear Mixed Model is defined as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\epsilon}$$

Where $\boldsymbol{\beta}$ are fixed effects (Rotation Treatment), $\mathbf{u}$ are random effects (Plot, Year), and $\boldsymbol{\epsilon}$ is the error term. These models are estimated using Restricted Maximum Likelihood (REML), which provides unbiased estimates of variance components in unbalanced designs.14

5.2 Modeling Temporal Autocorrelation

A central methodological challenge in LTE analysis is defining the covariance structure of the error term, $Var(\boldsymbol{\epsilon}) = \mathbf{R}$. Because soil has "memory" (nutrient carryover, seed banks), residuals from the same plot in adjacent years are highly correlated.

- **Compound Symmetry (CS):** Assumes constant correlation ρ between any two years. This is often too simplistic for long durations.
- **First-Order Autoregressive (AR1):** Assumes correlation decays exponentially with time lag: $\sigma^2 \rho^{|t_i - t_j|}$. This is biologically realistic for processes like nitrogen decay.
- **Spatial Power (SP(POW)):** A generalization of AR1 for unequally spaced time intervals.

Research analyzing LTE datasets consistently demonstrates that specifying the correct covariance structure is crucial to prevent Type I errors (false positives). A **Compound**

Symmetry with Heterogeneous Variances (CSH) structure is frequently cited as an optimal compromise, allowing for yield volatility to differ between years (e.g., drought vs. normal years) while maintaining a simplified correlation parameter.[14]

5.3 Stability Analysis and Variance Decomposition

Modern statistical analysis of rotations goes beyond mean yield comparisons to assess **Stability**. This involves modeling the Treatment × Year interaction variance. By partitioning this variance, analysts can determine which rotation systems are most resilient to climate variability. The "stability variance" (σ_i^2) for each rotation provides a metric of risk, essential for farmer decision-making.[14]

6. Machine Learning Architectures: RNNs and LSTMs

The limitations of Markovian models (limited memory) and econometric models (linearity assumptions) have driven the adoption of Deep Learning, specifically **Recurrent Neural Networks (RNNs)** and **Long Short-Term Memory (LSTM)** networks. These architectures are designed to model sequential data with long-term dependencies, making them ideal for analyzing decades of crop history.

6.1 The Vanishing Gradient and the LSTM Solution

Standard RNNs struggle to learn dependencies that span long time gaps (e.g., the effect of a legume planted 5 years ago) due to the "vanishing gradient" problem during backpropagation. LSTMs address this via a specialized cell architecture containing three gates:

1. Forget Gate (f_t): Determines what proportion of the previous cell state C_{t-1} to retain.

   $$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

   Biologically, this mimics the decay of soil nutrients or the mortality of weed seeds over time.
2. **Input Gate (i_t):** Decides what new information (current weather/crop) to store in the cell state.
3. **Output Gate (o_t):** Controls the information flow to the next hidden state h_t.[17]

This "gated" memory allows LSTMs to effectively model the **path-dependency** of agricultural systems, capturing complex rotation cycles (e.g., 5-year complex rotations) that HOMCs cannot parameterize efficiently.[19]

6.2 Hybrid Architectures: CNN-GAT-LSTM

Crop rotation is a spatio-temporal process. To capture both dimensions simultaneously, advanced methodologies employ hybrid architectures. A prominent example is the **CNN-GAT-LSTM** framework [20]:

- **Spatial Feature Extraction (CNN):** Convolutional Neural Networks process inputs like

weather maps or soil grids to extract local spatial features (intra-county or intra-field).

- **Relational Modeling (GAT): Graph Attention Networks** model the region as a graph where counties/fields are nodes and adjacency defines edges. The GAT mechanism learns weights for neighbors, allowing the model to understand that a drought in County A implies high risk for County B, but perhaps less for County C due to irrigation differences.
- **Temporal Integration (LSTM):** The spatially enriched features are then fed into an LSTM sequence to model the time evolution.

Ablation Studies: Research validating this architecture demonstrates that removing the LSTM component results in a significant drop in prediction accuracy (RMSE increases), confirming that temporal crop rotation information is as critical as spatial environmental data.[20]

7. The Transformer Revolution: Attention Mechanisms

The most recent and profound shift in methodology is the application of **Transformer** architectures, originally developed for Natural Language Processing (NLP), to crop rotation analysis. This approach conceptualizes crop sequences as a "language," where crops are words and rotations are sentences.

7.1 The "Crop-Language" Paradigm and BERT

In this paradigm, deep learning models are trained to understand the "syntax" of agriculture.

- **Pre-training via Masking:** Similar to **BERT (Bidirectional Encoder Representations from Transformers)**, models are pre-trained using Masked Language Modeling (MLM). A sequence might be presented as: Corn -> -> Wheat -> Soy. The model learns to predict the masked crop based on the bidirectional context. This forces the model to learn agronomic rules (e.g., "Soybeans usually follow Corn") without explicit programming.[21]
- Self-Attention Mechanism: The core innovation is the attention mechanism:

  $$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

  Unlike LSTMs, which process data sequentially, the Attention mechanism allows the model to "look" at the entire history simultaneously. It calculates a weight for every past year relative to the current prediction.
  - *Insight:* This is particularly powerful for capturing **irregular impacts**. For example, a severe drought event 4 years ago might have a massive impact on current soil structure. An LSTM might dilute this information over 4 time steps, but an Attention mechanism can assign a high weight directly to that specific year's token.[23]

7.2 Architectures: Cropformer and Temporal Encoders

Specific implementations like **Cropformer** utilize this architecture to classify crop types from

time-series satellite data.

- **Positional Encoding:** Since Transformers have no inherent sense of order (unlike RNNs), sinusoidal positional encodings are added to the input embeddings to represent the sequence order (Year 1, Year 2, etc.).[25]
- **Multi-Head Attention:** This allows the model to focus on different types of relationships simultaneously. One "head" might focus on short-term rotation (last year's crop), while another head focuses on long-term phenological trends (climate shift over 10 years).

Comparative studies indicate that Transformer-based models (e.g., SITS-BERT, Cropformer) achieve superior classification accuracy compared to RNNs and Random Forests, particularly in "Few-Shot" learning scenarios where labeled training data is scarce. They effectively transfer learned representations from one region to another.[21]

8. Generative AI and Large Language Models (LLMs)

The frontier of rotation methodology lies in **Generative AI**. By fine-tuning Large Language Models (LLMs) like GPT-4 or specialized variants (AgriGPT) on vast corpora of agronomic text and structured crop history data, researchers are developing systems that can *generate* optimal rotation schedules.

- **Recommendation Systems:** Instead of just predicting *what* a farmer will plant (classification), these models act as decision support systems, suggesting *what should* be planted to maximize yield or sustainability.
- **Text-to-Rotation:** Farmers can interact with these models using natural language ("I have a corn nematode problem and sandy soil, suggest a 5-year rotation"). The model leverages its internal "world model" of agronomy—learned during training—to output a symbolic sequence that adheres to biological constraints.[27]
- **Accuracy:** Experiments fine-tuning GPT-2 on crop rotation tabular data converted to text descriptions have reported accuracy rates exceeding 99% in specific recommendation tasks, outperforming traditional ML models by leveraging the massive pre-trained knowledge base of the LLM.[27]

9. Comparative Analysis and Performance Metrics

Evaluating the success of these diverse methodologies requires standardized metrics. The field relies heavily on the **Confusion Matrix** and derived statistics.

9.1 Key Metrics

- **Overall Accuracy (OA):** The proportion of correctly classified pixels/fields.
- Kappa Coefficient (κ): A robust measure that accounts for the possibility of the agreement occurring by chance. This is crucial in crop mapping where one class (e.g., Corn) might dominate the landscape.

  $$\kappa = \frac{p_o - p_e}{1 - p_e}$$

Where p_o is the observed agreement and p_e is the expected agreement by chance.

- **F1-Score:** The harmonic mean of Precision and Recall, vital for evaluating rare crops in a rotation (e.g., a specific cover crop).

9.2 Methodological Comparison

The following table synthesizes the performance and characteristics of the primary methodologies discussed [21]:

Methodology	Primary Mechanism	Temporal Memory	Spatial Context	Typical Accuracy (OA / Kappa)	Best Use Case
Markov Chain (1st Order)	Probabilistic Transition	Short (t-1)	Low (unless Spatial Markov)	Moderate	Regional Simulation, Steady State Analysis
High-Order Markov (HOMC)	Multi-step Probability	Medium (t-m)	Low	Good (but suffers sparsity)	Complex Rotation Simulation
Random Forest (RF)	Ensemble Decision Trees	Feature Engineered	Hand-crafted features	High (~0.95 / ~0.90)	Static Classification, General Baselines
CNN-LSTM	Convolutions + Recurrence	Long (Gated)	High (Local convolution)	Very High	Spatio-temporal Yield Prediction
Transformer (Cropformer)	Self-Attention	Global / Infinite	High (if Vision Transformer)	Superior (>0.97 / >0.95)	Large-scale Mapping, Few-shot Learning

Table 2: Comparative overview of rotation analysis methodologies. Accuracy values are indicative based on reviewed literature for crop type mapping tasks.

10. Conclusion and Future Outlook

The methodology of rotation analysis has traversed a trajectory from **descriptive statistics** to **predictive intelligence**.

1. **Stochastic Models** (Markov Chains) remain the gold standard for understanding the *structure* of agricultural systems and testing for equilibrium/stationarity. They provide the transparent, interpretable probabilities required for policy analysis.
2. **Econometric Models** (MNL/DCM) provide the *causal* link to human behavior, allowing

researchers to simulate how pricing and policy shift rotation choices.

3. **Deep Learning** (LSTM/Transformers) has solved the *prediction* problem. By treating crop history as a complex language and the landscape as a connected graph, these models achieve unprecedented accuracy in forecasting land use and yield.

Future Directions: The convergence of these fields is imminent. **Physics-Guided Neural Networks (PGNN)** are emerging as a hybrid methodology, embedding the biological constraints of LTEs and the economic logic of Utility theory directly into the loss functions of Transformers. This will create "Digital Twins" of agricultural systems that are not only accurate in prediction but consistent with the laws of agronomy and economics. Furthermore, the integration of **Generative AI** promises to shift the field from *analyzing* rotations to *optimizing* them, proactively designing sustainable landscapes for a changing climate.

Works cited

1. An Assessment of Crop-Specific Land Cover Predictions Using High-Order Markov Chains and Deep Neural Networks, accessed January 16, 2026, https://www.nass.usda.gov/Education_and_Outreach/Reports,_Presentations_and_Conferences/Journal_Articles/Assessment%20of%20Crop%20Specific%20Land%20Cover%20Preditctions%20Using%20High%20Order%20Markov%20Chains%20and%20NN.pdf
2. Identifying representative crop rotation patterns and grassland loss in the US Western Corn Belt | Request PDF - ResearchGate, accessed January 16, 2026, https://www.researchgate.net/publication/265469918_Identifying_representative_crop_rotation_patterns_and_grassland_loss_in_the_US_Western_Corn_Belt
3. Pre-season crop type mapping using deep neural networks | Request PDF - ResearchGate, accessed January 16, 2026, https://www.researchgate.net/publication/343365043_Pre-season_crop_type_mapping_using_deep_neural_networks
4. A multifaceted approach combining frequency-domain image co-registration, transformer-based parcel segmentation, and Bi-LSTM for crop classification - Research journals - PLOS, accessed January 16, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0299350
5. A Price Risk Early Warning Model for Agricultural Products Based on the Integration of Deep Learning and Knowledge Graphs - IEEE Xplore, accessed January 16, 2026, https://ieeexplore.ieee.org/iel8/6287639/10820123/11164274.pdf
6. Recovering Forecast Distributions of Crop Composition: Method and ..., accessed January 16, 2026, https://www.mdpi.com/2071-1050/12/7/2917
7. Farmers and Habits: The Challenge of Identifying the Sources of Persistence in Tillage Decisions Steven Wallander, USDA Economic, accessed January 16, 2026, https://www.aeaweb.org/conference/2018/preliminary/paper/4esS2eYd
8. Discrete-Time Markov Chain Method for Predicting Probability of Crop Yield Variability, accessed January 16, 2026, https://www.mdpi.com/2673-4834/6/4/142
9. Predictive Soil Health Economic Calculator (P-SHEC) - Farmland Information Center, accessed January 16, 2026, https://farmlandinfo.org/wp-content/uploads/sites/2/2025/11/pshec-technical-ma

nual.pdf
10. Spatiotemporal evolution regional differences and decoupling effects of greenhouse gas emissions from animal husbandry in Henan Province - PubMed Central, accessed January 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12583758/
11. Modeling acreage decisions within the multinomial Logit framework - IDEAS/RePEc, accessed January 16, 2026, https://ideas.repec.org/p/ags/inrasl/211011.html
12. Modeling acreage decisions within the multinomial Logit framework - This document is discoverable and free to researchers across the globe due to the work of AgEcon Search. Help ensure our sustain, accessed January 16, 2026, https://ageconsearch.umn.edu/record/211011/files/WP%20SMART-LERECO%2009-17.pdf
13. On the economic theory of crop rotations: value of the crop rotation effects and implications on acreage choice modeling - IDEAS/RePEc, accessed January 16, 2026, https://ideas.repec.org/p/ags/inrasl/205299.html
14. Long-term experiments with cropping systems: case studies ... - IRIS, accessed January 16, 2026, https://iris.uniss.it/bitstream/11388/84479/5/LTE_Stat_rev2_.pdf
15. The Design and Analysis of Long-Term Rotation Experiments - ResearchGate, accessed January 16, 2026, https://www.researchgate.net/publication/270633985_The_Design_and_Analysis_of_Long-Term_Rotation_Experiments
16. Building ANOVA-models for long-term experiments in agriculture - Fixing the bridge between biologists and statisticians, accessed January 16, 2026, https://www.statforbiology.com/2020/stat_lte_modelbuilding/
17. Deep Learning-Based Crop Prediction Using LSTM And GRU For Sustainable Agriculture, accessed January 16, 2026, https://www.researchgate.net/publication/393084735_Deep_Learning-Based_Crop_Prediction_Using_LSTM_And_GRU_For_Sustainable_Agriculture
18. Integrating multi-modal remote sensing, deep learning, and attention mechanisms for yield prediction in plant breeding experiments - Frontiers, accessed January 16, 2026, https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1408047/full
19. Methodology for multi-temporal prediction of crop rotations using recurrent neural networks - PolyPublie, accessed January 16, 2026, https://publications.polymtl.ca/54344/1/2023_Dupuis_Methodology_Multi-temporal_Prediction_Crop_Rotations.pdf
20. A Temporal–Geospatial Deep Learning Framework for Crop Yield ..., accessed January 16, 2026, https://www.mdpi.com/2079-9292/13/21/4273
21. Cropformer: A new generalized deep learning classification approach for multi-scenario crop classification - PMC - PubMed Central, accessed January 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10017990/
22. End-to-End Transformer-Based Models in Textual-Based NLP - MDPI, accessed January 16, 2026, https://www.mdpi.com/2673-2688/4/1/4

23. Robust and timely within-season conterminous United States crop ..., accessed January 16, 2026, https://www.nass.usda.gov/Research_and_Science/Cropland/docs/In-Season%20Crop%20Mapping%20Transformer-compressed.pdf
24. Investigating attention mechanisms for plant disease identification in challenging environments - PMC - NIH, accessed January 16, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11066637/
25. Vision Transformers in Precision Agriculture: A Comprehensive Survey - arXiv, accessed January 16, 2026, https://arxiv.org/html/2504.21706v4
26. Full article: Solution for crop classification in regions with limited labeled samples: deep learning and transfer learning, accessed January 16, 2026, https://www.tandfonline.com/doi/full/10.1080/15481603.2024.2387393
27. (PDF) CROP RECOMMENDATION SYSTEM USING MACHINE LEARNING - ResearchGate, accessed January 16, 2026, https://www.researchgate.net/publication/382318196_CROP_RECOMMENDATION_SYSTEM_USING_MACHINE_LEARNING
28. Agriculture Meets AI: Precision Crop Planning and Sustainable Farming Using GenAI - Evnek Quest, accessed January 16, 2026, https://www.evnekquest.com/post/agriculture-meets-ai-precision-crop-planning-and-sustainable-farming-using-genai
29. Land Use/Land Cover Factor Values and Accuracy Assessment Using a GIS and Remote Sensing in the Case of the Quashay Watershed in, accessed January 16, 2026, https://d-nb.info/1258811170/34
30. High Resolution Crop Type and Rotation Mapping in Farming–Pastoral Ecotone in China Using Multi-Satellite Imagery and Google Earth Engine - MDPI, accessed January 16, 2026, https://www.mdpi.com/2072-4292/17/10/1707
31. Can a history of crop rotations improve the prediction of soil organic carbon in the Andes? integrating machine learning multi-a - bioRxiv, accessed January 16, 2026, https://www.biorxiv.org/content/10.64898/2025.12.06.692759v1.full.pdf